

Problem 3

Evaluate the integral:

$$I = \int \frac{(x+1)^2}{\sqrt{x}} dx.$$

Solution

To solve the given integral, we first expand and simplify the integrand, and then integrate term by term.

Step 1: Expand the numerator

The numerator is $(x+1)^2$, which expands as:

$$(x+1)^2 = x^2 + 2x + 1.$$

Thus, the integrand becomes:

$$\frac{(x+1)^2}{\sqrt{x}} = \frac{x^2}{\sqrt{x}} + \frac{2x}{\sqrt{x}} + \frac{1}{\sqrt{x}}.$$

Simplify each term:

$$\frac{x^2}{\sqrt{x}} = x^{3/2}, \quad \frac{2x}{\sqrt{x}} = 2x^{1/2}, \quad \frac{1}{\sqrt{x}} = x^{-1/2}.$$

Therefore, the integral becomes:

$$I = \int x^{3/2} dx + 2 \int x^{1/2} dx + \int x^{-1/2} dx.$$

Step 2: Integrate each term

Using the power rule for integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \text{for } n \neq -1,$$

we compute each term:

1. **First term: $\int x^{3/2} dx$:

$$\int x^{3/2} dx = \frac{x^{3/2+1}}{3/2+1} = \frac{x^{5/2}}{5/2} = \frac{2}{5}x^{5/2}.$$

2. **Second term: $2 \int x^{1/2} dx$:

$$\int x^{1/2} dx = \frac{x^{1/2+1}}{1/2+1} = \frac{x^{3/2}}{3/2} = \frac{2}{3}x^{3/2}.$$

Multiply by 2:

$$2 \int x^{1/2} dx = 2 \cdot \frac{2}{3}x^{3/2} = \frac{4}{3}x^{3/2}.$$

3. **Third term: $\int x^{-1/2} dx$:

$$\int x^{-1/2} dx = \frac{x^{-1/2+1}}{-1/2+1} = \frac{x^{1/2}}{1/2} = 2x^{1/2}.$$

Step 3: Combine the results

Combine all three terms:

$$I = \frac{2}{5}x^{5/2} + \frac{4}{3}x^{3/2} + 2x^{1/2} + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = \frac{2}{5}x^{5/2} + \frac{4}{3}x^{3/2} + 2x^{1/2} + C.$$